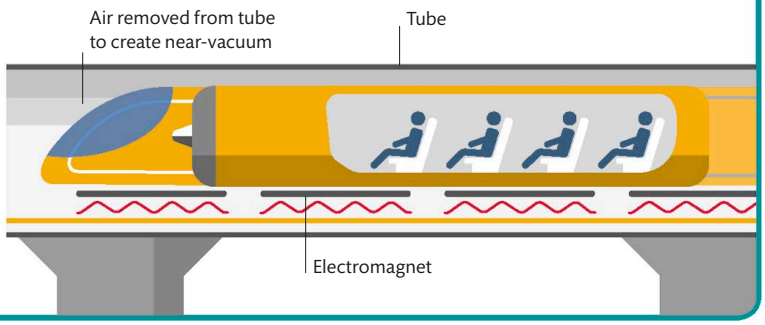
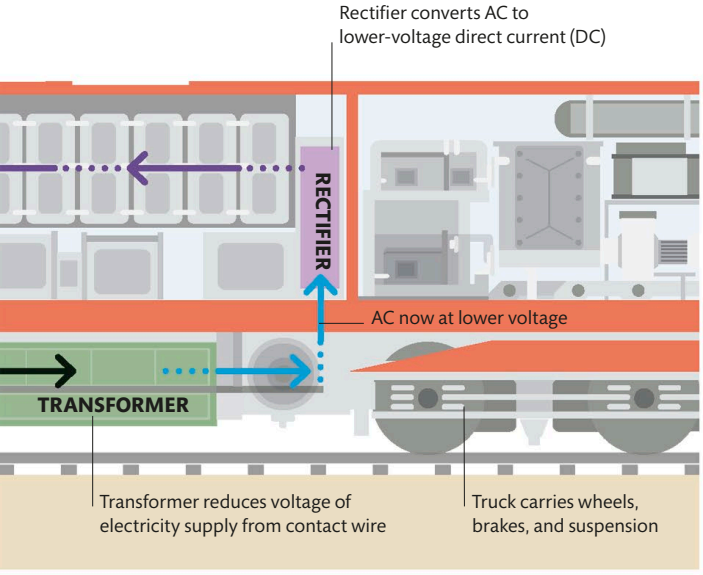
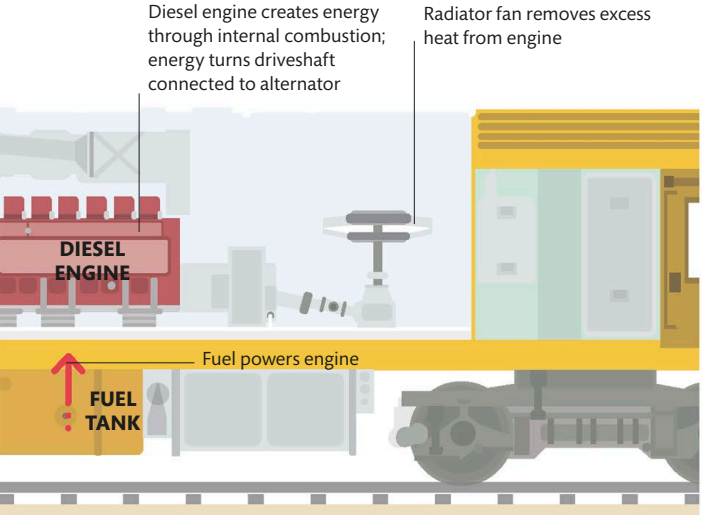
HYPERLOOP

A Hyperloop is an experimental train designed to travel faster than a jet airliner. Passenger pods travel inside a tube that is a near-vacuum. Air is removed to reduce the piston effect (a buildup of air in front of the train) and to enable the pods to travel faster by reducing friction. Electromagnets beneath the train and on the track repel or attract each other to generate lift and thrust.



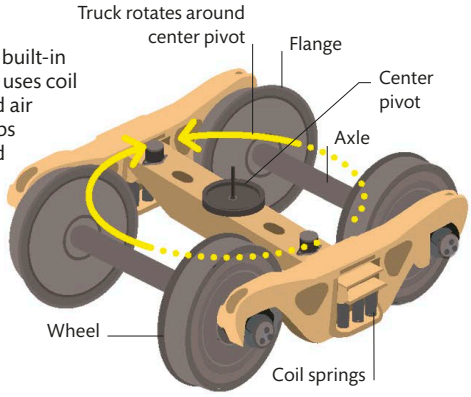


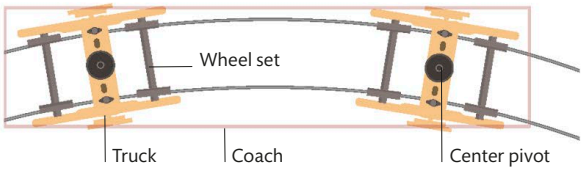


Railroad trucks and wheels

Every section of a train is supported at each end by a frame system, or railroad truck, in which the wheel sets (axles and wheels) are mounted. Some trucks can turn to follow bends in the track. The wheels are made of solid steel and run on steel rails to minimize rolling friction. Each wheel has a projecting rim, or flange, on one side that helps to hold it on the rails.

Smoothing the ride
A railroad truck has a built-in suspension system. It uses coil springs, dampers, and air bags to soak up bumps and vibrations caused by uneven tracks. The wheels stay in contact with the rails, while the locomotive and coaches above move along smoothly.





Turning bends
A long train with steel wheel sets traveling along steel rails is inherently rigid. To allow trains to follow bends, some modern railroad trucks have a built-in steering mechanism, with a steering beam and levers hinged around a center pivot, which allows the wheel sets to turn.